

Active visual continuous seam tracking based on adaptive feature detection and particle filter tracking

RONG FAN,¹ PENG ZHANG,^{2,*} FENGYUN GUO,³ JIE RONG,² AND XUPENG LIAN¹

¹College of Electronic Information and Optical Engineering, Taiyuan University of Technology, Taiyuan, 030024, China

²College of Physics, Taiyuan University of Technology, Taiyuan, 030024, China

³Zhongyi New Material Intelligent Manufacturing Research Institute, Taiyuan 030024, China

*zhangpengphysics@163.com

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Welding seam tracking based on online programming is the future trend of intelligent production. However, most of the existing image processing methods have certain limitations in the adaptability, accuracy, and robustness of weld feature point detection. The online welding method of gas metal arc welding (GMAW) based on active vision sensing is studied in this paper. The Steger sub-pixel detection method is used to guarantee the accuracy of feature point extraction, and a self-adaptive search window and self-adaptive slope extraction are proposed on this basis. The self-adaptive window is generated according to the linear information of the weld area, and the scale factor and range threshold constraint are added to realize the real-time detection of the weld feature information. Screening the center pixel of the laser stripe in the self-adaptive window of the current frame by the initial slope or the self-adaptive slope of the previous frame, the linear information of the weld area is obtained. The self-adaptive slope of the current frame is fitted by the random sampling consistency method, and the pixel margin is retained to adapt to the linear detection of different continuous welds. When arc light and other serious interference make it difficult to obtain weld information, a particle filter is used to make the best prediction of the weld position. Finally, the welding robot platform based on laser vision sensing was built to test various continuous welds of the butt weld, fillet weld, and lap weld. Experimental results show that the detection speed is 27 ms, and the accuracy of detection and tracking can respectively reach 0.03 mm and 0.78 mm, which meets the requirements of weld detection and tracking. © 2024 Optica Publishing Group

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1. INTRODUCTION

Weld tracking technology is a metal connection technology. By using the intelligent welding robot to develop modern manufacturing [1], the consistency of welding quality and the continuity of weld work can be ensured. At present, the control mode of welding robots mostly is teaching programming. The welding robot is reproduced through manual position calibration [2]. But there is some human subjective error in this operation mode, and welding can only be completed at a specific position in a specific occasion space [3]. Offline programming makes the welding robot simulate the real working state by building a CAD model [4], but there are certain application errors between the actual environment and the simulation environment [5]. However, the autonomous programming method overcomes the shortcomings of the above control methods [6]. The welding robot is highly adaptable to extracting all kinds of information in the welding environment by various sensors [7], without manual planning of the working path.

The non-contact measurement method is used to detect the seam information and state change of the welded workpiece in real time by a visual sensor, which provides reliable visual guidance information for subsequent tracking technology [8]. Since the auxiliary laser source has a high energy density, it has a positive effect on the accuracy of automatic welding [9]. The active vision sensing method with a laser as the auxiliary source is adopted widely. It irradiates the specific shape of the laser onto the surface of the object to be measured to form a stable laser stripe or structured light stripe by a line laser or structured light laser, which can achieve accurate detection of the plane and curved welds, as well as stable tracking of narrow butt joints and welds in narrow spaces [10]. However, in the dynamic welding process of laser vision sensing, the welding-related residual stress and cumulative thermal deformation will affect the accurate tracking of the weld position, declining the welding quality. Moreover, the laser vision weld feature extraction method cannot cope with different material weldments, and the adaptability of the application space is limited [11]. In addition, the image

processing method is the important process of active tracking [12]. Zhang *et al.* [13] applied the laser sensor to measure and track the groove weld with the proposed visual four-step method to improve the positioning accuracy of straight and curved welds. Kiddee *et al.* [14] took the intersection region of adaptive fitting lines as a template, using template matching to extract the weld edge for real-time weld tracking. Jia *et al.* [15] used an improved Kalman filter algorithm based on space-time information to extract weld feature points in the welding interference environment, flexibly tracking the welding seam combined with the welding gun control strategy. He *et al.* [16] extracted the weld contour from the strong arc background by the clustering method according to the visual attention features of the weld, removing the interference information in the image to improve the anti-interference ability. Xiao *et al.* [17] trained the Faster R-CNN model based on the feature information of continuous and discontinuous welds, and a self-adaptive weld feature extraction method is proposed to realize automatic welding of various typical weld seam tracking.

Although the current study of image processing methods for the active vision sensor to extract weld information proposed some improvement measures, it has some limitations in real-time, accuracy, and robustness. This paper presents an improved method of weld detection and tracking. The self-adaptive search window is introduced into the Steger sub-pixel detection algorithm, constantly updating the detection window to reduce the detection time of subsequent welding points. The self-adaptive slope is constantly updated on the laser fringe center line fitted by the random sample consensus method (RANSAC) in the self-adaptive window, and different types of continuous weld information can also be obtained according to their self-adaptive slopes. Weld information is easily influenced by interference factors such as arc splashing, smoke diffusion, and plate defects during dynamic welding [18], making it difficult to obtain laser fringes clearly and accurately. The optimal estimation of the particle filter is proposed for the weld image, which can't be accurately detected by the Steger algorithm to predict the

weld feature points. The Steger sub-pixel detection method is combined with the particle filter tracking algorithm to track the weld path stably and reliably. The detection performance of weld feature points is verified on different continuous weld data sets captured by the laser vision sensor, which shows the detection accuracy and adaptability of the applied algorithm. The tracking performance of real-time welding points of the continuous weld path is analyzed in the welding robot system to further test the accuracy and robustness.

2. APPLICATION OF LASER VISION

In the dynamic welding process, the real-time existence of environmental interference factors brings difficulties in obtaining the weld center point accurately. This paper proposes the improved method of seam tracking based on laser vision to solve these problems. It can improve the detection speed of extracting seam information on this basis and can also detect and track various continuous welds. Autonomous welding mainly has five parts, including laser vision image acquisition, laser vision calibration, laser stripe weld point detection, weld trajectory tracking, and control planning strategy. The workflow of autonomous welding is shown in Fig. 1.

A. Laser Vision Model

The laser vision sensor is one of the autonomous sensing methods for intelligent welding of welding robots. The active welding sensor is designed and built in this paper, which has the functions of measure and locating the weld position. The sensor is equipped with an industrial camera, laser transmitter, filter glass sheet, dustproof glass sheet, and their respective fixing brackets.

The sensor model is shown in Fig. 2. The relative angle between the camera and laser transmitter is 40° . The camera optical axis and the laser are in a state of coincidence in the working distance so that the laser is in the camera's visual field during the welding process. The detection accuracy of the camera at the optimal welding working distance of 150 mm is 0.01 mm. The

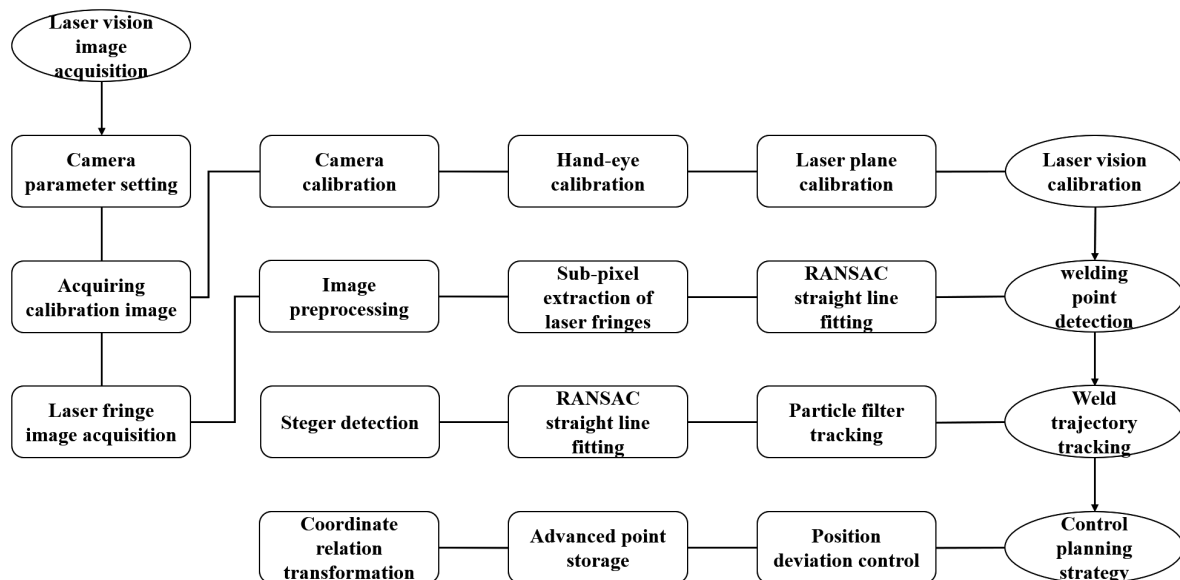


Fig. 1. Work flow chart of autonomous welding.

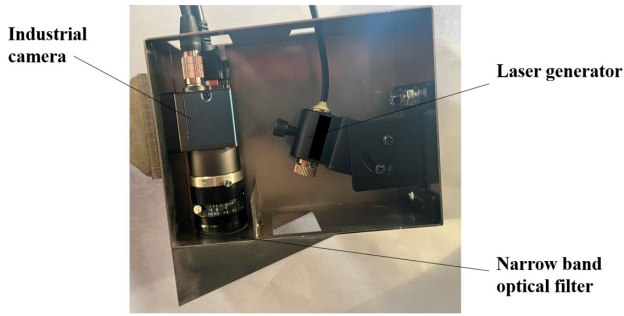


Fig. 2. Physical diagram of the active vision sensor.

Table 1. Parameters of the Laser Vision Sensor

Main Equipment	Equipment Parameter	Parameter Value
Camera (MER2-160-75GM)	Sensor type	Sony IMX273 global shutter CMOS
	Pixel size	$3.45 \times 3.45 \mu\text{m}$
	Image size	1440×1080
	Frame rate	75.6 fps
Camera lens	Resolution ratio	2 million pixels
	Focal distance	16 mm
	Field of view	$46 \times 34 \text{ mm}$
Line laser transmitter	Laser power	200 mw
	Line width	1 mm
	Wave length	660 nm

main equipment parameters of the laser vision sensor are shown in Table 1.

B. Laser Vision Calibration

Laser vision calibration is the premise of robot welding and effects the control accuracy of guided welding [19]. The robot visual guidance relationship is obtained by laser vision calibration, which is shown in Fig. 3. The calibration method and subsequent image processing algorithm adopted in this paper are realized by VS2015 with C++ as the development language, which provides preconditions for realizing robot autonomous positioning and online welding.

Since the image detection speed does not match the seam tracking speed of GMAW welding, the camera is paraxially fixed in front of the longitudinal coordinate of the welding torch with

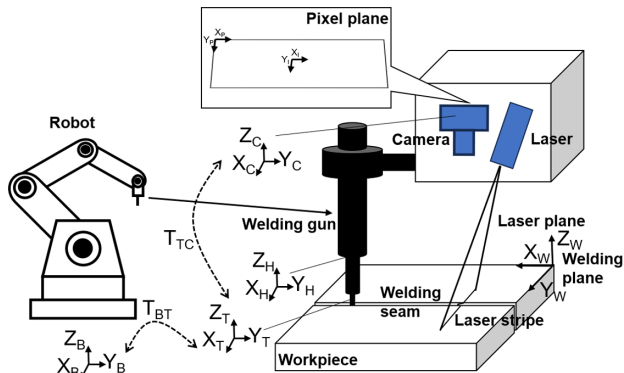


Fig. 3. Coordinate system diagram of the welding robot and laser vision sensor.

a prepositive distance of 80 mm, which can provide sufficient time for image acquisition and processing. The essence of camera calibration is to solve the camera parameters in the imaging process, reflecting the mapping relationship between the pixel image and physical world expressed by Eq. (1):

$$\begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix} = \frac{1}{z_c} \begin{bmatrix} f_x & 0 & x_0 \\ 0 & f_y & y_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R & T \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}, \quad (1)$$

Where (x_p, y_p) is the two-dimensional coordinate of the pixel point on the image. (x_w, y_w, z_w) is the corresponding world coordinate point. z_c is the scale factor. (f_x, f_y) is the horizontal and vertical scaling factor of the pixel. (x_0, y_0) is the pixel coordinate of the image origin point. R is the rotation matrix. T is the translation matrix. The camera internal parameter and external parameter are obtained by Zhang's calibration method [20]. The calibration results of the specific internal parameter matrix and radial distortion coefficients are shown in Table 2.

Based on the fixed calibration plate, the hand-eye transformation matrix is calculated after multiple welding torch movements and different camera poses to obtain the real-time feedback expression of continuous welding [21]. The hand-eye transformation matrix is obtained by using Eq. (2):

$$\begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \\ z_t \\ 1 \end{bmatrix}, \quad (2)$$

where (x_c, y_c, z_c) is the three-dimensional coordinate of camera point. (x_t, y_t, z_t) is the three-dimensional coordinate of corresponding tool point. R is the 3×3 rotation matrix. T is the 3×1 translation matrix. The detailed calibration result of the above-mentioned matrices is shown in Table 2.

Because of the pinhole imaging principle, the weld image collected by the camera is two-dimensional plane information. The laser plane calibration is needed to complete weld point information to realize the three-dimensional shape measurement of the laser fringe [22]. The laser plane calibration is completed by solving parameters of the following laser plane Eq. (3):

$$z_c = ax_c + by_c + c, \quad (3)$$

Table 2. Calibration Result of Vision System

Calibration Type	Calibration Result
Internal calibration	$\begin{bmatrix} 5607.3090 & 0 & 842.2524 \\ 0 & 5631.3397 & 624.62468 \\ 0 & 0 & 1 \end{bmatrix}$
Distortion coefficient	$[-0.5610 \quad 0.9562 \quad -0.0047 \quad -0.0107]$
Hand-eye calibration	$\begin{bmatrix} 0.9861 & -0.0080 & 0.1656 & 43.0912 \\ 0.0321 & 0.9891 & -0.1435 & 11.7794 \\ -0.1626 & 0.1468 & 0.9756 & -60.6310 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Laser plane calibration	$[-0.0006 \quad -0.0212 \quad 0.0048]$

Where (x_c, y_c, z_c) is the camera coordinate of laser point. a , b , and c are plane parameters, which are shown in Table 2.

C. Laser Fringe Image Processing

Through the active vision continuous seam tracking algorithm combining adaptive Steger detection and particle filter tracking, the feature points of the visual seam are extracted and tracked. The improved image processing algorithm mainly includes image preprocessing, laser stripe centerline extraction, line fitting to obtain feature point, and welding feature point tracking, which can improve the positioning accuracy of the weld starting point, real-time detection of the weld path, and robustness of the weld trajectory tracking in automatic welding process. Moreover, the improved method has certain universality for continuous weld detection and does not need to establish the a priori model [23]. The improved image processing method is shown in Fig. 4.

1. Image Preprocessing

The visual image is readily affected by welding factors such as arc splashing, smoke diffusion, and plate defects in the welding environment [24], resulting in the phenomenon of illumination change, similar background interference, target occlusion, or loss in the obtained laser stripe image, which brings certain difficulties to the subsequent extraction of weld information of the laser stripe. Therefore, this paper preprocesses the obtained laser fringe image, which mainly includes graying, extracting ROI

region, linear morphological operation, median filtering, gray pixel binarization, and area filtering. The specific preprocessing flow is shown in Fig. 5.

The proposed preprocessing method is applied to obtain laser fringes in the process of weld path detection, and the detection effect is shown in Fig. 6. In this paper, a linear structure with the size of 1×5 is used to process the ROI region of the gray image by linear morphological operation, and the contour information of the laser fringe is highlighted. Since the pixel gray value and occupied area of the light stripe area in the ROI area of the laser stripe image are relatively large, gray pixel binarization is used to keep the area with the large gray value in the gray image, including light bar areas, which eliminates the interference of dark areas. At the same time, the area of the contour region is calculated by extracting the connected region with the large gray value, and the smaller bright region is removed by the area threshold. Clear laser fringe images are obtained by the above preprocessing method, most interference factors are eliminated, and only the ROI region is preprocessed to improve the algorithm efficiency, which ensures the accurate detection of the subsequent weld feature points.

2. Laser Stripe Centerline Extraction

Since the laser fringe is not easily disturbed by external noise, the Steger detection method applied extracts the centerline of the laser fringe with sub-pixel accuracy in the local field of welding view. The laser fringe image is ridged in three-dimensional

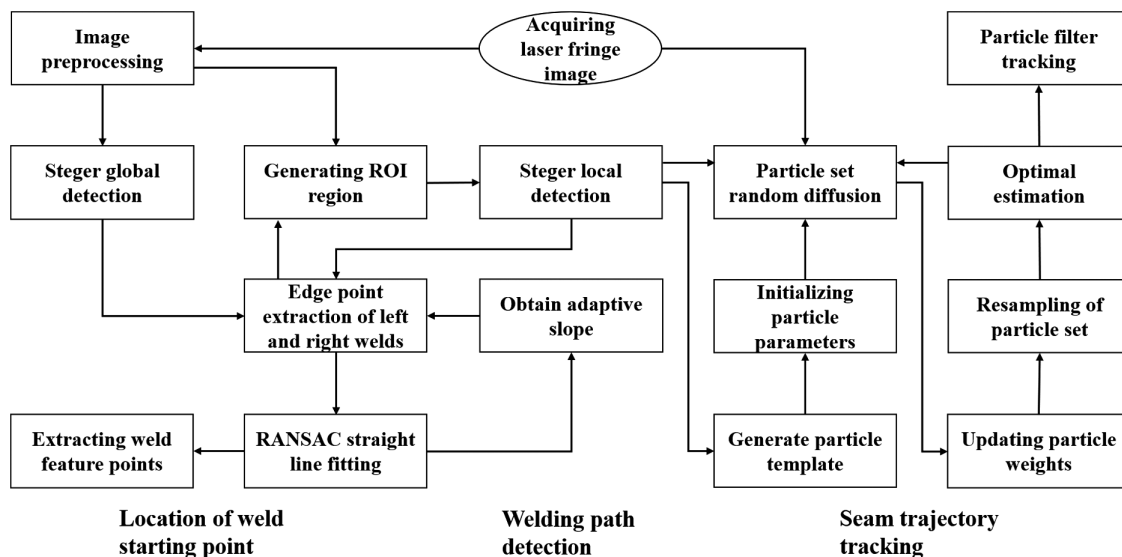


Fig. 4. Flowchart of the improved image processing algorithm.

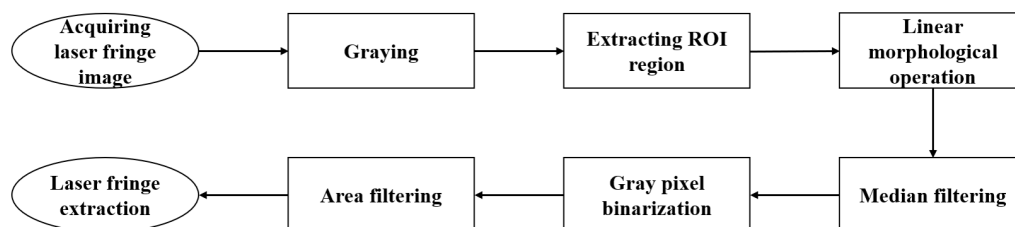


Fig. 5. Flowchart of image preprocessing.

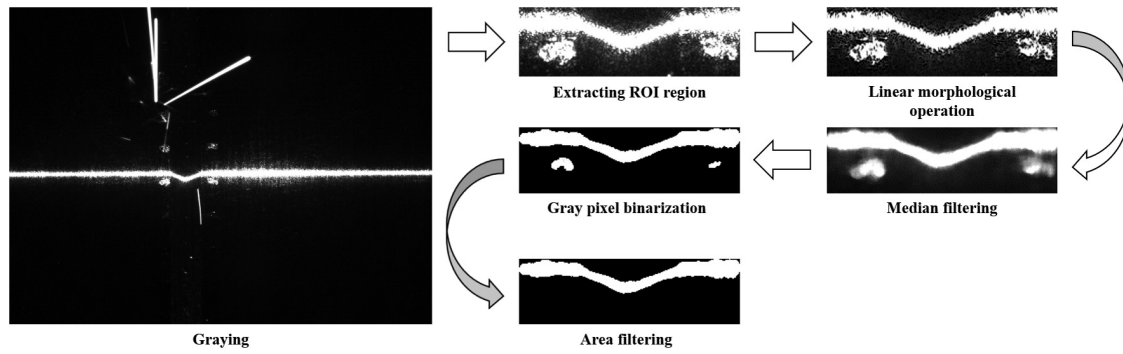


Fig. 6. Effect diagram of weld path detection.

space. The light intensity gradient along the normal direction changes greatly, while the light intensity gradient along the tangent direction changes relatively little. The Steger method calculates the maximum gradient direction of the Gaussian filtered image by using the Hessian matrix, which can be expressed by Eq. (4):

$$H(x, y) = \begin{bmatrix} h''_{xx} & h''_{xy} \\ h''_{xy} & h''_{yy} \end{bmatrix}, \quad (4)$$

where (x, y) is the pixel point of the Gaussian filtering image. h'_x is the first pixel derivative along the x direction. h'_y is the first pixel derivative along the y direction. h''_{xx} is the continuous second-order pixel derivative along the x direction. h''_{xy} is the pixel derivative along the x direction and then along the y direction. h''_{yy} is the continuous second-order pixel derivative along the y direction.

The laser centerline is obtained by pixel convolution operation, and the coordinates (p_x, p_y) of the laser centerline are obtained by Eq. (5), which can achieve sub-pixel accuracy,

$$(p_x, p_y) = (x_0 + tn_x, y_0 + tn_y), \quad (5)$$

where (x_0, y_0) is the pixel point on the light stripe. (n_x, n_y) is the normal direction. t is the vector modulus in the normal direction, and it can be calculated by the Taylor expansion Eq. (6):

$$t = -\frac{h'_x n_x + h'_y n_y}{h''_{xx} n_x^2 + 2h''_{xy} n_x n_y + h''_{yy} n_y^2}. \quad (6)$$

When the laser centerline meets Eq. (7), the point is the sub-pixel center point on the laser centerline,

$$(x_0 + tn_x, y_0 + tn_y) \in ([x_0 - 0.5, x_0 + 0.5], [y_0 - 0.5, y_0 + 0.5]), |tn_x|, |tn_y| \in [0, 0.5]. \quad (7)$$

The Steger algorithm is applied to search the self-adaptive window in the preprocessed laser fringe image, and the feature points of the largest normal gradient is extracted. However, in addition to the laser stripe area, there may be large arc areas in the larger bright area, which seriously affects the accurate extraction of the laser centerline. It is necessary to post-process the laser fringe image to ensure the accurate identification of the light centerline. The post-processing flow is shown in Fig. 7, which

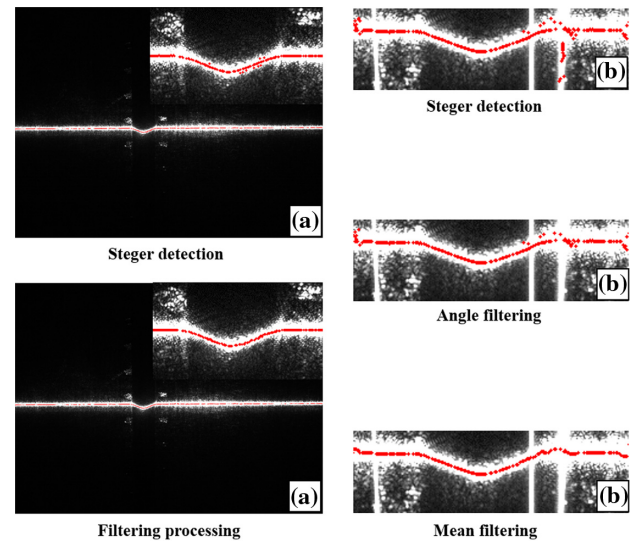


Fig. 7. Post-processing flow of the laser center line. (a) Effect diagram of global detection. (b) Effect diagram of local detection.

fully reflects the robustness of the self-adaptive Steger algorithm by combining directional filtering and mean filtering.

The left and right weld edge points in the laser centerline are extracted according to the initial given slope or the modified detection slope, and the self-adaptive ROI region for subsequent frame image detection is generated. The a priori model of continuous welds is not considered in the detecting process. Only the left and right slopes of the laser center point are used to select the point to be measured, and the weld center point is determined by the screening point with the maximum y coordinate. The left and right weld edge points can be generated to avoid the existence of Steger random detection error. The self-adaptive ROI region is introduced in this paper, and it is expressed by Eq. (8). Applying the self-adaptive window restricts the extraction number of the laser center points. However, due to the fewer edge points detected, this leads to the continuous convergence of ROI areas in subsequent frames. Therefore, the ROI threshold is set in this paper to adjust the window size adaptively,

$$\text{ROI} = [x_0 - w/s \quad y_0 - h/s \quad sw \quad sh], \quad (8)$$

where (x_0, y_0) is the pixel point extracted from the upper left edge points of the current frame image. w is the maximum distance extracted from the left and right edge points of the

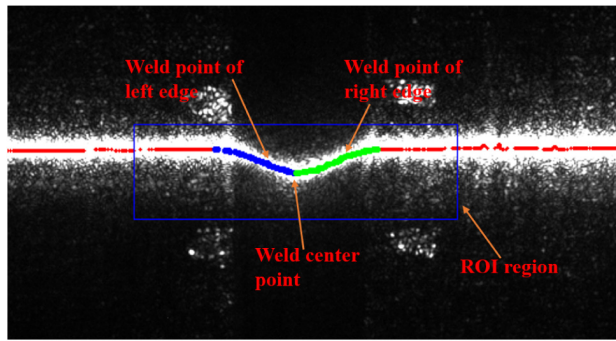


Fig. 8. Setting of the ROI region.

current frame image along the x direction. h is the maximum distance extracted from the upper and lower edge points of the current frame image along the y direction. s is the ROI size factor, defined as 2. $(x_0 - w/s, y_0 - h/s)$ is the starting point of the current frame ROI area. $(s w, s h)$ is the width and height of the current frame ROI area. The ROI area is set as shown in Fig. 8.

3. Line Fitting to Obtain Feature Point

In the self-adaptive ROI region, the welding center point detected by the Steger method is directly used as the weld tracking point. There may be some accidental errors in the single detection information, which requires the full extraction of the sub-pixel center point set. The left and right fitting lines are carried out by using the left and right weld edge point set. By comparing the fitting effects of the RANSAC and least square method, it is found that the RANSAC method has higher fitting accuracy. The laser stripe has a certain width, and the laser shows irregular reflection at the weld, which makes the centerline points extracted by Steger algorithm messy and dense. There may be “bad points” at the weld edge points, which affects the accuracy of line fitting.

The welding edge points are extracted by the given initial detection slope, and the intersecting two fitting lines are obtained with the RANSAC algorithm. However, different types of continuous welds have different detection slopes. Only specific initial slopes and correction slopes cannot meet the detection of different continuous welds. A certain number of margins are retained by the left and right edge weld points to achieve the generation of RANSAC fitting lines, which ensures the accurate detection of different continuous weld images and improves the adaptability of the Steger algorithm. The detection effect of the RANSAC fitting algorithm based on the self-adaptive slope is shown in Fig. 9. The self-adaptive slope is constantly updated to correct the detection slope in the subsequent weld detection. The intersection point of the fitting lines is calculated by the two straight line equations obtained in parallel to obtain the welding point of the current frame. The self-adaptive slope k is expressed by Eq. (9):

$$k = a (|k_1| + |k_2|) / 2, \quad (9)$$

where k_1 is the slope of the left weld fitting line. k_2 is the slope of the right weld fitting line. a is the correction constant defined as 0.9 and 4.5, which are used as the correction constants of the

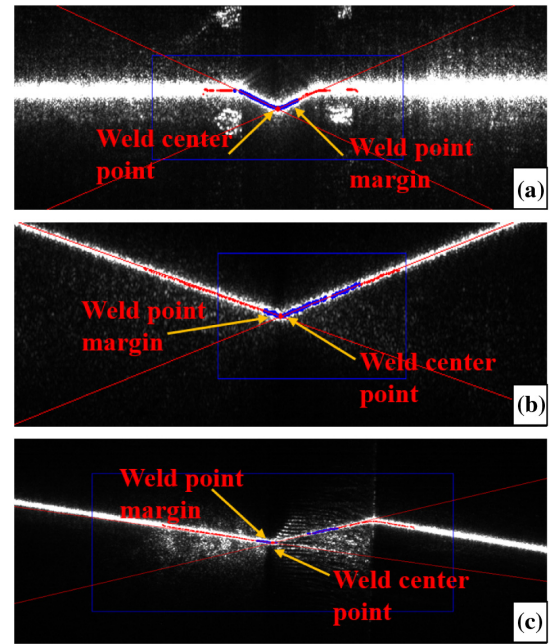


Fig. 9. Straight line fitting effect of different welds. (a) Fitting result of butt weld. (b) Fitting result of fillet weld. (c) Fitting result of lap weld.

detection slope and angle filtering, respectively. The initial value of k is defined as 0.2.

4. Welding Feature Point Tracking

The detection of seam information is mainly interfered by the arc splash in the actual environment. The adaptive Steger method alone can't extract the weld information in real-time image frames. The existence of similar backgrounds can easily affect the detection of moving targets, resulting in tracking offset and even making the target occluded or lost so that the tracking fails. It is necessary to combine the particle filter tracking algorithm to make the optimal prediction.

According to the self-adaptive Steger method, the laser stripe image with relatively few interference factors is accurately detected from the current frame image. The established self-adaptive window is used as the active region of the particle swarm in the next frame image. The slope and intercept of the extracted fitting lines are provided to generate the target template line. The left and right template lines are expressed as follows in Eq. (10):

$$y^{(1)} = k_0 x + b_0, \quad y^{(2)} = k_0 x + b_0. \quad (10)$$

In the next frame image, the particle swarm is initialized based on the slope k_0 and intercept b_0 of the template lines $y^{(1)}(k_0, b_0)$, $y^{(2)}(k_0, b_0)$, and 100 sets of left and right sample lines $y^{(1,2)}(k_i, b_i)$ with the same slope k_i and intercept b_i are generated, respectively. The position of the sample particles is in the detection position of the self-adaptive Steger in the previous frame image. The position distribution of the sample particle swarm is expressed by Eq. (11), and the weights of the sample lines are defined as $1/n$,

$$\{y^{(1,2)}(k_i, b_i) | k_i = k_0, b_i = b_0\}, \quad (i = 1, 2, 3 \dots, 100). \quad (11)$$

The predictive motion model Eq. (12) is used to diffuse the position state of the generated particle set in the ROI region. Based on the detection results in the previous frame image, the target positions are predicted by the transferred particles with different slopes and intercepts in the current frame image

$$y_t^{(1,2)}(k_i, b_i) = y_0^{(1,2)}(k_i, b_i) + 2(y_{t-1}^{(1,2)}(k_i, b_i) - y_0^{(1,2)}(k_i, b_i)) - (y_{t-2}^{(1,2)}(k_i, b_i) - y_0^{(1,2)}(k_i, b_i)) + V_t, \quad (12)$$

where $y_t^{(1,2)}$ is the current frame prediction value of sample lines. $y_0^{(1,2)}$ is the initial value of sample lines. $y_{t-1}^{(1,2)}$ is the previous frame observation value of sample lines. $y_{t-2}^{(1,2)}$ is the previous frame observation value of sample lines in the previous frame, and V_t is the system noise that obeys the normal distribution.

The left and right weld lines in the self-adaptive window are detected by the self-adaptive Steger algorithm. The feature information of the target template is established by the gray histogram of fitting lines. The weights of the transfer particles are updated by comparing the gray histogram $H_i(I)$ of sample lines with the gray histogram $H_0(I)$ of template lines. The weights of the transfer particles are used as the observation values to correct the prediction results of sample lines. The gray calculation range of sample lines is expressed by Eq. (13). The weight calculation of sample lines is expressed by Eq. (14),

$$\{(x, y) | y^{(1,2)}(k_i, b_i), x \in \text{ROI}, p(x, y) > 0\}, \quad (13)$$

$$w_i^{(1,2)} = 1 - \sqrt{1 - \frac{1}{\sqrt{\bar{H}_0 \bar{H}_i N^2}} \sum \sqrt{H_0(I) H_i(I)}}, \quad (14)$$

where (x, y) is the current frame image coordinate, ROI is the ROI area updated in the previous frame, $p(x, y)$ is the pixel value of the current frame image coordinate, $w_i^{(1,2)}$ is the weight

value of sample lines, $\bar{H}_0, \bar{H}_i$ is the histogram mean, and N is the histogram gray level.

After the particle weights are normalized, the line particles with larger weights are copied according to the probability distribution of random sampling, and the line particles with smaller weights are removed, but the particle number of sample lines is constant. The sample particles with smaller weights are less than the sample particles with larger weights to correct the position and weight of left and right linear particles so that the particle set always converges to the target line. The weight normalization $\hat{w}_i^{(1,2)}$ is expressed by Eq. (15):

$$\hat{w}_i^{(1,2)} = \frac{w_i^{(1,2)}}{\sum_{i=1}^{100} w_i^{(1,2)}}, \quad \sum_{i=1}^{100} \hat{w}_i^{(1,2)} = 1. \quad (15)$$

The retained sampling particle weights are equalized in the optimal particle state prediction. The optimal slope and intercept estimations obtained by equalized sample particles are used to obtain the best sample lines in the current frame image. The welding center point is obtained by intersecting the left and right lines of the optimal estimation. The optimal estimation of the welding edge lines is realized by Eq. (16):

$$\hat{y}^{(1,2)}(k, b) = \sum_{i=1}^{100} \hat{w}_i^{(1,2)} y^{(1,2)}(k_i, b_i). \quad (16)$$

In the subsequent interference image frames, the sampling slope and intercept of linear particle sets in the current frame image are used as the motion reference information of the transfer particles. The ROI region is updated according to the position of the tracking point obtained by the optimal estimation of the current frame image, which is used as the active region of linear particle sets in the next frame image. When the image noise is reduced, the self-adaptive Steger algorithm is applied for real-time detection of the weld information to ensure the continuity of the tracking target weld point. The weld point extraction process based on self-adaptive Steger detection and particle filter tracking is shown in Fig. 10, and

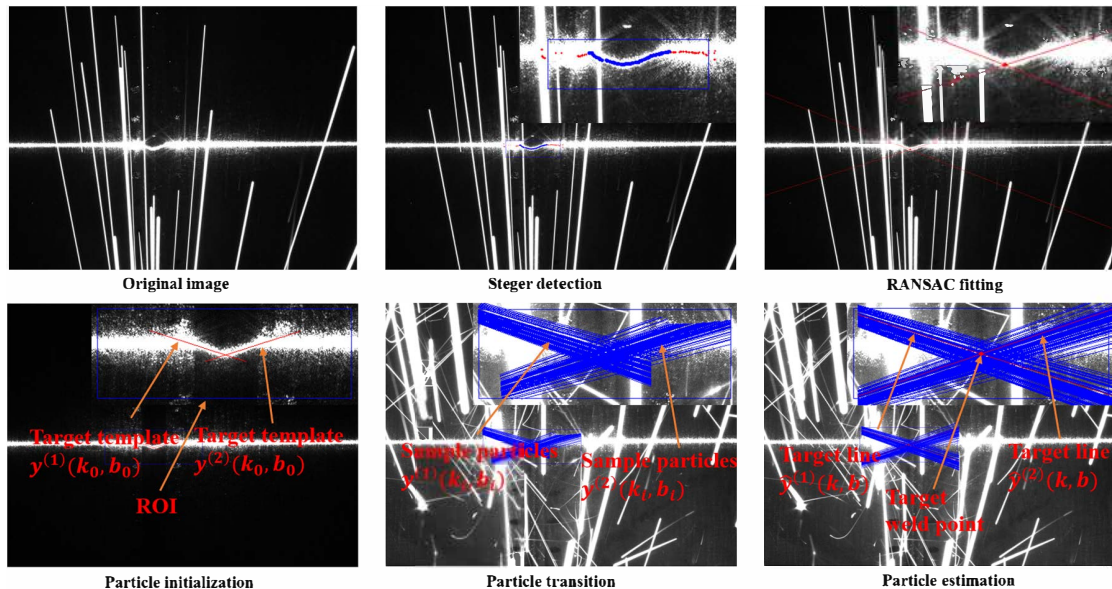


Fig. 10. Application of improved method in butt welding detection.

the best prediction strategy of the particle filter is marked and explained.

3. EXPERIMENTAL ANALYSIS

The improved method is verified by an active vision sensing welding robot system. The continuous weld images of the butt groove, fillet groove, and lap groove in the welding process are captured by the laser vision sensor. The extraction accuracy and detection speed of the feature information are simulated and tested on the collected weld data set. In addition, the continuous weld seam are tracked in real time by the welding robot. The actual welding process is analyzed with specific welding parameters, and we compare the weld positioning results of teaching programming and self-programming online.

A. Welding Robot System

The welding robot system based on active vision sensing mainly includes the laser vision part, welding part, and weldments. The laser vision system mainly consists of a linear laser light source emitter, Da Heng industrial camera, and Da Heng lens. The welding system is mainly equipped with a YRC1000 control cabinet, D350SW welding power supply, Kaierda six-axis manipulator, welding torch, and wire feeding mechanism. The main system structure is shown in Fig. 11. According to the coordinate deviation between the current welding position point and the extracted feature position point, the weld tracking trajectory is continuously calibrated. The welding robot is used

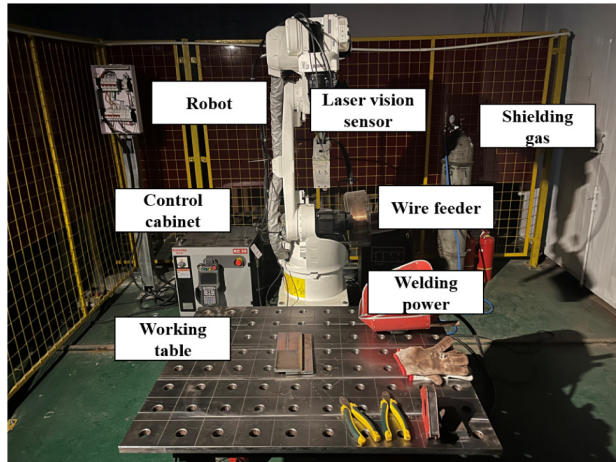


Fig. 11. Intelligent welding robot system diagram.

to weld the workpiece stably and dynamically according to the optimized welding parameters to complete the autonomous welding tracking process. Specific welding process parameters and workpiece dimensions are shown in Table 3.

B. Image Detection Analysis

The current weld point is manually marked on the laser stripe image obtained, taking the manually marked point as the image reference point of the actual weld. By comparing the welding points detected by the classical Steger algorithm and the proposed improved algorithm with the image marked points respectively, the recorded detection errors of the butt weld, fillet weld, and lap weld are shown in Table 4. The average error of the basic algorithm is 3.58 pixels (114.56 μm), and the standard deviation is 4.46 pixels (142.72 μm). The average error of the improved algorithm is 0.80 pixels (25.60 μm), and the standard deviation is 0.83 pixels (26.56 μm). Compared with the basic algorithm, the detection result of the improved algorithm is better, which is due to the poor robustness of the basic algorithm and the random error of the single detection information. The average detection speeds of the basic algorithm and the improved algorithm are 135 ms and 27 ms, respectively, and the time used by the improved algorithm is reduced by 80%, which meets the real-time continuous tracking of the weld joint.

Through the simulation test of extracting weld feature information on different types of weld data sets, the detection and analysis results are shown in Fig. 12. The self-adaptive Steger detection algorithm can be used to extract weld sub-pixel points in the normal process of weld point location to ensure accurate detection of weld information. When there is relatively strong arc light, smoke, or welding plate reflection, the self-adaptive Steger algorithm can also fully identify the center points of various continuous welds in the adaptive window with adaptive slope, which has certain anti-interference ability. The self-adaptive Steger algorithm provides accurate template information for particle tracking when very strong arc noise interference occurs. The particle filter algorithm can accurately predict the left and right edge weld lines in subsequent image frames according to real-time template information and then extract the welding center point. Taking the optimal estimated target point as the tracking position of the welding point where it is difficult for the Steger method to extract weld information, the continuous weld is predicted and tracked, which realizes the continuous tracking of different continuous welds, and the robustness and adaptability of the detection performance are improved.

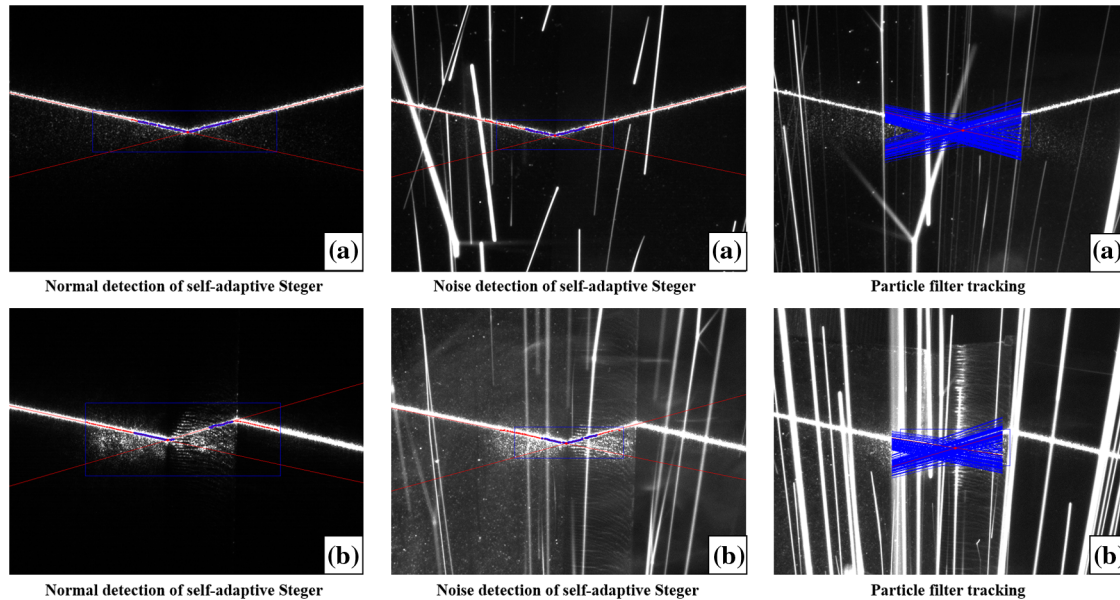
Table 3. Relevant Welding Process Parameters of Experimental Application

Welding Parameters	Given Value	Welding Parameters	Given Value
Material type	Q235 steel plate	Welding method	GMAW
Groove type	Butt, fillet, and lap	Pulse frequency	125 HZ
Work piece	200 × 100 × 5 mm	Welding current	150 A
Weld length	200 mm	Welding voltage	22 V
Weld thickness	5 mm	Welding speed	6 mm/s
Wire type	THQ-50C	Shielding gas	80%Ar + 20%CO ₂
Wire diameter	1.2 mm	Gas-flow rate	18 L/min
Dry elongation	15 mm	Wire feeding rate	5.1 m/min

Table 4. Detection Error Statistics of Different Welds^a

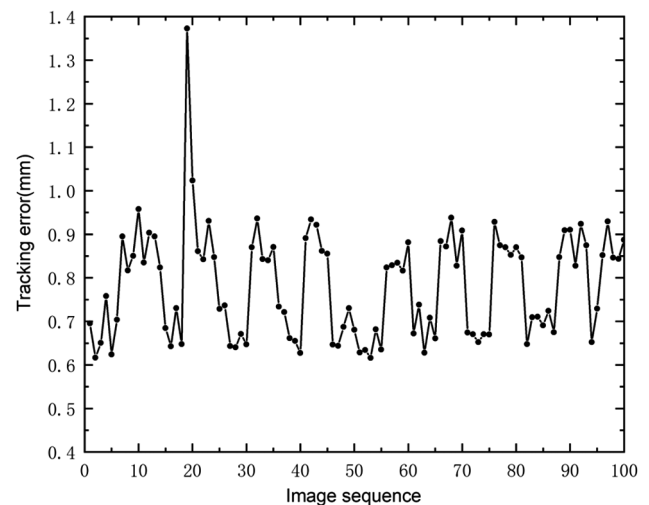
Applied Method	Seam Type (MAE/Pixel)	Mean Absolute Error (RMSE/Pixel)	Root Mean Square Error (MAXE/Pixel)	Max Error (MINE/Pixel)	Min Error
Steger	Butt groove	2.957	3.453	8.106	0.694
	Fillet groove	2.869	3.555	11.052	0.006
	Lap groove	4.918	5.975	14.928	0.760
Proposed	Butt groove	0.590	1.165	1.57	0.043
	Fillet groove	0.848	1.378	3.269	0.164
	Lap groove	0.983	1.468	3.613	0.297

^aMAE, mean absolute error; RMSE, root mean square error; MAXE, max error; MINE, min error.

**Fig. 12.** Detection effect of different welds. (a) Detection result of fillet weld. (b) Detection result of lap weld.

C. Welding Tracking Analysis

For purpose of further testing the tracking performance of the improved method, 100 sample points were selected on the butt, fillet, and lap welds with a distance of 200 mm, and the spacing was 2 mm. The sampling error analysis was carried out based on the teaching mark points. The three-dimensional coordinates of the sample points are obtained by teaching positioning and online calibration. The Euclidean distance between the teaching mark points and the predicted calibration points is used as the tracking deviation information. The error distribution of the sampling points is shown in Figs. 13–15. The average tracking error of the butt weld is 0.78 mm, and the maximum tracking error is 1.37 mm. The average tracking error of the fillet weld is 0.50 mm, and the maximum tracking error is 1.16 mm. The average tracking error of the lap weld is 0.62 mm, and the maximum tracking error is 1.22 mm. The existence of tracking error may be caused by factors such as coordinate calibration, mechanical motion, and workpiece assembly. In general, the average tracking error of the welding point is 0.63 mm, and the maximum tracking error is 1.37 mm. According to the feedback of real-time weld information, the online programming can generate the tracking trajectory independently, which meets the requirements of actual welding accuracy and robustness.

**Fig. 13.** Tracking accuracy of the butt welding seam.

4. CONCLUSION

In this paper, a continuous seam tracking method based on active vision sensing is proposed, and the practical significance is realized on the applied welding robot. The main advantages are as follows: (1) Through the adaptive improvement of the

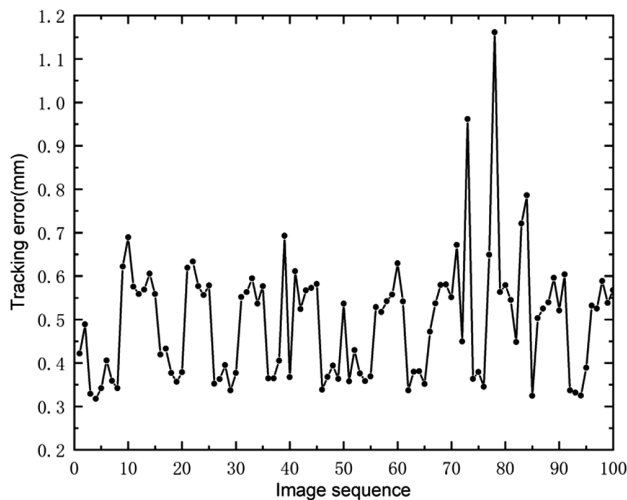


Fig. 14. Tracking accuracy of the fillet welding seam.

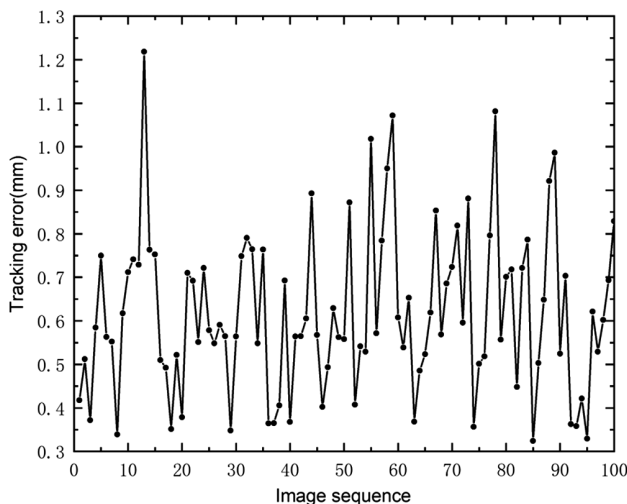


Fig. 15. Tracking accuracy of the lap welding seam.

traditional Steger algorithm, not only can the weld information be fully extracted with sub-pixel accuracy from the self-adaptive window of the laser stripe image but also the laser centerline can be accurately extracted on the laser stripe image to reduce the influence of interference factors in the image, which has certain anti-interference ability. (2) Many kinds of weld detection methods need to establish an a priori model to detect all kinds of welds, which is not flexible. The RANSAC algorithm based on the self-adaptive slope is adopted to generate their own adaptive slopes according to the weld characteristics of various continuous welds, which is used for accurate detection of all kinds of welds and has certain universality. (3) In real-time seam tracking, the image quality is easily interfered by the external environment such as arc light, smoke, and weld plate reflection, which makes the laser stripe image unable to be clearly obtained. The particle filter algorithm is applied to predict and identify the seam image that the Steger algorithm cannot accurately detect, which ensures the robustness of welding seam trajectory tracking on the basis of meeting the real-time requirements.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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